

Design, Synthesis and Biological Evaluation of Benzimidazole-pyridine-Piperidine Hybrids as a New Class of Potent Antimicrobial Agents

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Abstract: A series of novel benzimidazole-pyridine-piperidine hybrids was synthesized in good yields and characterized by spectral and elemental analyses. The compounds **4a-h** and **5a-c** were evaluated for their in vitro antibacterial activity against gram-positive organisms viz., *Bacillus subtilis*, *Staphylococcus aureus*, *Staphylococcus epidermidis*, gram-negative organisms viz., *Escherichia coli*, *Pseudomonas aeruginosa*, *Klebsiella pneumonia* and also against fungal strains like *Candida albicans*, *Saccharomyces cerevisiae* of yeasts, *Aspergillus flavus*, *Aspergillus niger* according to the CLSI Standard Protocol. Compound **5a** showed promising activity against all tested organism excluding *Bacillus subtilis* when compared to standard drugs.

Keywords: 2-substituted benzimidazole, antibacterial activity, antifungal activity, ethyl-piperidine-4-carboxylic acid, hybrid molecules, pyridine.

1. INTRODUCTION

The treatment of infectious diseases remains an important issue because of the combination of factors including emerging newer infectious diseases and increasing number of multi-drug resist antimicrobial pathogens. This problem is particularly pronounced for the Gram-positive bacteria [1-4]. Despite a large number of antibiotics and chemotherapeutics available for medical use, the emerging resistance to old and new antibiotics has created a substantial need for new class of antimicrobial agents. A potential approach to overcome the problem of antibiotic resistance is to design innovative agents with different modes of action so that no cross resistance with present drugs can occur [5].

Benzimidazoles are a familiar group of heterocyclic compounds possessing a wide variety of biological activity. Benzimidazole and its derivatives are reported to be physiologically and pharmacologically active and find applications in the treatment of several diseases like epilepsy, diabetes, antifertility [6, 7]. It is an important pharmacophore and privileged structure in medicinal chemistry [8, 9] encompassing a diverse range of biological activities including antibacterial [10], anti-inflammatory, analgesic [11, 12], antihistamine [13], antiulcerative [14], antioxidant [15], antiproliferative [16, 17], antiallergic [18, 19], antitumor [20, 21], antikinase [22, 23], anticancer [24-28], cytotoxic [29], and anti-HIV activities [30, 31]. On the other hand, Pyridines are

considered privileged scaffolds in drug discovery with a wide array of biological activities. The pyridine nucleus is prevalent in numerous natural products and is extremely important in the chemistry of biological systems. Pyridine derivatives have been used as bactericides [32], fungicides [33], and anticancer agents [34-37]. Moreover, Piperidine and its analogues, an important pharmacophore of many drug molecules, are reported to act as antibacterials [38], antitubercular agents [39] and AChE inhibitors [40].

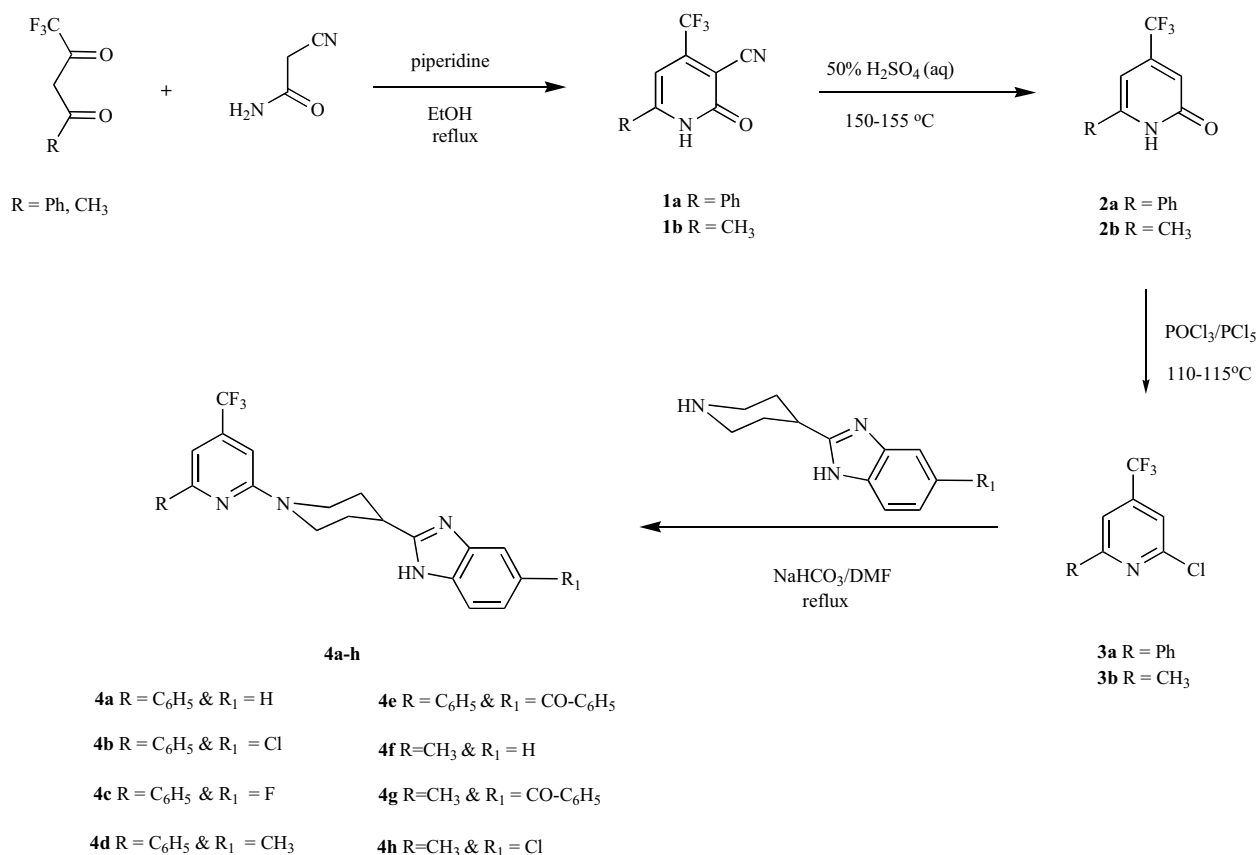
In view of the above mentioned findings, and as a continuation of our effort to identify new candidates that may be of value in designing new, potent, selective, and less toxic biologically active molecules [41-45], we report herein the synthesis and antimicrobial evaluation of some novel structural hybrids incorporating the benzimidazole, pyridine and piperidine ring systems. This combination was suggested to investigate the influence of such hybridization and structure variation on the anticipated biological activities, hoping to add some synergistic biological significance to the target molecules. The target compounds were rationalized so as to comprise some pharmacophores that are believed to be responsible for the biological activity of some relevant chemotherapeutic agents. The substitution pattern of the benzimidazole and pyridine rings was carefully selected in order to confer different electronic environment to the molecules.

2. RESULTS AND DISCUSSION

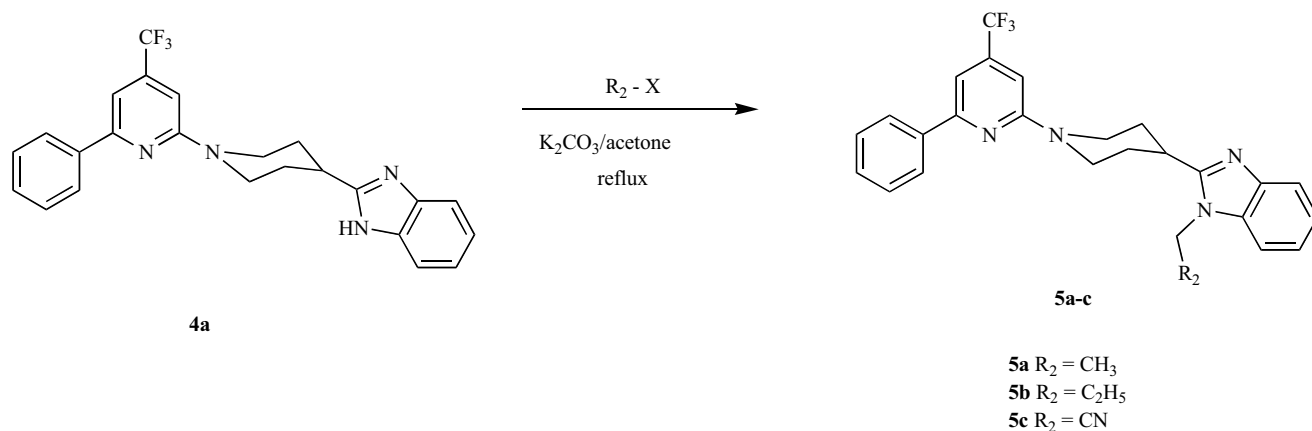
2.1. Chemistry

The general route for the synthesis of the target benzimidazole-pyridine-piperidine hybrids is depicted in Scheme (1).

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Scheme (1). Synthesis of benzimidazole-pyridine-piperidine hybrid molecules **4a-h**.



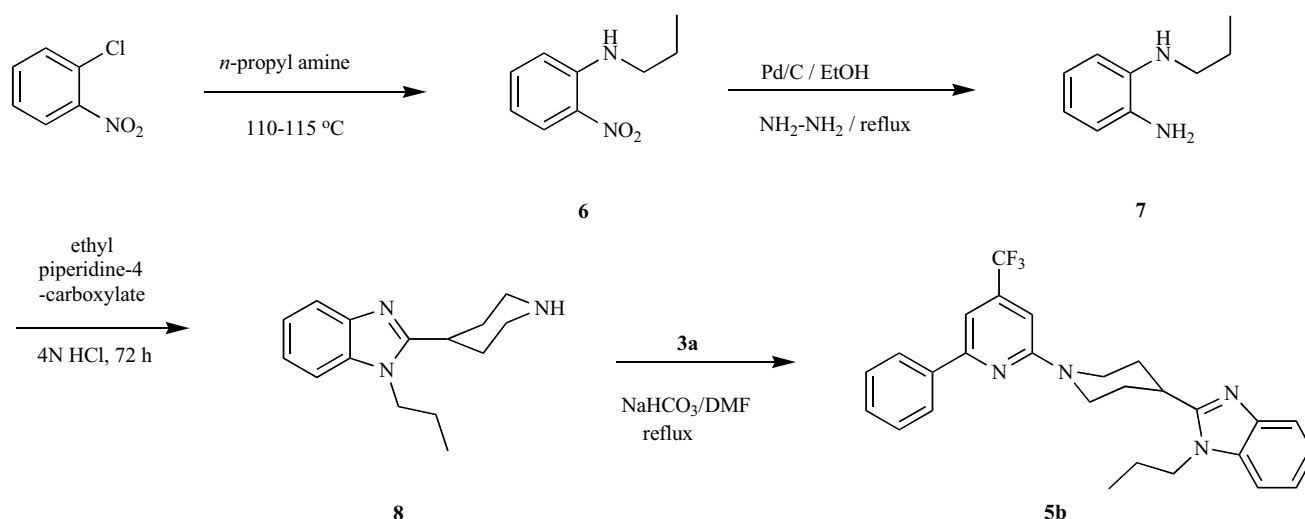
Scheme (2). Synthesis of hybrid molecules **5a-c** from **4a**.

The synthetic methodology was designed to start with the reaction of cyano acetamide and 4,4,4-trifluoro-1-methyl/phenylbutane-1,3-dione in presence of base at 75-80 °C that afforded 2-oxo-6-methyl/phenyl-4-(trifluoromethyl)-1,2-dihydropyridine-3-carbonitrile **1a-b** [46]. The pyridone was further hydrolyzed and decarboxylated to give 6-methyl/phenyl-4-(trifluoromethyl) pyridin-2(1*H*)-one **2a-b** [47] in excellent yields. The pyridones **2a-b** when reacted with freshly distilled POCl₃ in the presence of PCl₅ furnished 2-chloro-6-methyl/phenyl-4-(trifluoromethyl) pyridine **3a-b**. The condensation of 2-chloro-6-methyl/phenyl-4-(trifluoromethyl) pyridine with various 2-(piperidin-4-yl)-1*H*-benzo[*d*]imidazoles [48] in presence of NaHCO₃/DMF furnished 2-(1-(6-methyl/phenyl-4-(trifluoromethyl) pyridin-2-yl) piperidin-4-

yl)-1*H*-benzo[*d*]imidazoles **4a-h** in excellent yields and the schematic diagram is depicted in (Scheme 1).

The compound **4a** was reacted with alkylhalide in K₂CO₃/acetone under refluxing conditions furnished 1-substituted-2-(1-(6-phenyl-4-(trifluoromethyl) pyridin-2-yl) piperidin-4-yl)-1*H*-benzo[*d*]imidazole **5a-c** (Scheme 2). The reaction was initially conducted in dry DMF solvent at refluxing temperature that furnished the products **5a-c** in 40 % yield. The product yield was improved to 72-83 % by changing the solvent to acetone.

The structure of compound **5b** was established by synthesizing an independent route starting from 1-chloro-2-nitrobenzene reacting with *n*-propyl amine at 110-115 °C



Scheme (3). Synthesis of 2-(1-(6-Phenyl-4-(trifluoromethyl) pyridin-2-yl) piperidin-4-yl)-1-propyl-1*H*-benzo[*d*] imidazole **5b**.

Table 1. *In vitro* antibacterial activity of hybrid molecules **4a-h**, **5a-c**.

Compound	MIC (µg/mL)					
	<i>B.subtilis</i>	<i>S.aureus</i>	<i>S.epidermidis</i>	<i>E.coli</i>	<i>P.aeruginosa</i>	<i>K.pneumoniae</i>
4a	>150	>150	>150	>150	>150	>150
4b	>150	>150	>150	>150	>150	>150
4c	>150	>150	>150	>150	>150	>150
4d	>150	>150	>150	>150	>150	>150
4e	>150	>150	>150	>150	>150	>150
4f	>150	>150	75	>150	>150	>150
4g	>150	>150	75	>150	>150	>150
4h	150	>150	75	75	75	>150
5a	150	9.375	4.68	9.375	4.68	37.5
5b	>150	>150	>150	>150	>150	>150
5c	>150	>150	>150	>150	>150	>150
Penicillin	1.562	1.562	3.125	12.5	12.5	6.25
Norfloxacin	9.375	1.17	1.17	18.75	1.17	9.375

DMSO-negative control showed no activity.

condition affording 2-nitro-*N*-propylaniline **6**. The later was treated with metal catalyzed reduction using Pd/C in the presence of hydrazine hydrate resulting in *N*1-propylbenzene-1, 2-diamine **7**. Compound **7** was reacted with ethyl-piperidine-4-carboxylic acid [48] furnishing 2-(piperidin-4-yl)-1-propyl-1*H*-benzo[*d*]imidazole **8**. Compound **8** was further reacted with 2-chloro-6-phenyl-4-(trifluoromethyl) pyridine affording 2-(1-(6-phenyl-4-(trifluoromethyl) pyridin-2-yl) piperidin-4-yl)-1-propyl-1*H*-benzo[*d*]imidazole **5b** (Scheme 3). The structure of alkylated compound **5b** was found to be identical with the compound obtained by unambiguous route (Scheme 3) in all respects which was further confirmed by super imposable IR spectra.

All the final compounds were well characterized with spectral and elemental analysis.

2.2. Evaluation of Biological Activity

2.2.1. In Vitro Antibacterial Activity

The title hybrid molecules **4a-h** and **5a-c** were tested against bacterial organism such as Gram-positive organisms *Bacillus subtilis* (MTCC 441), *Staphylococcus aureus* (MTCC 96), *Staphylococcus epidermidis*, Gram-negative organisms viz *Escherichia coli* (MTCC 443), *Pseudomonas aeruginosa* (MTCC 741), *Klebsiella pneumonia* (MTCC

Table-2. *In vitro* antifungal activity of 1-Ethyl-2- (1- (6-phenyl-4- (trifluoromethyl) pyridine-2- yl) piperidin-4-yl) -1*H*-benzo[d]imidazole **5a**.

Compound	Zone of Inhibition in mm							
	<i>C. albicans</i>		<i>S. cerevisiae</i>		<i>A. niger</i>		<i>A. flavus</i>	
	100µg	150µg	100µg	150µg	100µg	150µg	100µg	150µg
5a	7	10	7	10	-	-	-	-
Amphotericin(50 µg)	23.5		22		24		25	

Negative control; DMSO showed no activity; well or cup method; the concentrations are expressed in mg/mL and inhibitory zone diameters are expressed in mm.

618) and also against fungal strains like *Candida albicans* (MTCC 227), *Saccharomyces cerevisiae* (MTCC36) of yeasts, *Aspergillus flavus* (MTCC 277), *Aspergillus niger* (MTCC 282) was done according to CLSI Standard Protocol [49] and the results were tabulated in Table 1, 2. Compound **5a** showed excellent activity against all tested organism excluding *Bacillus subtilis* (MTCC 441). In gram-negative organisms, it is more active against *Escherichia coli* (MTCC 443) strain MIC value 9.375 with comparison to norfloxacin which is a broad range microbial agent and also penicillin. It is also active against gram-positive organisms with comparison to penicillin especially *Staphylococcus epidermidis* with MIC value 4.68. Compounds **4f-h** showed moderate activity against bacterial organisms. The optimum alkyl group on benzimidazole NH is found to be as ethyl moiety and did not show any interesting activity either by increasing the number of carbons or by replacement of nitrile groups. The standard samples are considered as resistant to *Escherichia coli* (MTCC 443), however compound **5a** showed enhanced activity than standard samples.

2.2.2. In Vitro Antifungal Activity

The *in vitro* antifungal activities of hybrid molecules **4a-h** and **5a-c** were evaluated against four fungi including *Candida albicans* Y0109, *Candida albicans* SC5314, *Saccharomyces cerevisiae*, *Aspergillus niger*, *Aspergillus flavus*. The positive control, Amphotericin, was obtained from their respective manufacturers. Compound **5a** showed moderate activity against fungal organism like *Candida albicans* (MTCC 227), *Saccharomyces cerevisiae* (MTCC36). Remaining compounds showed poor activity against all fungal organisms. Compound **5a** is identified as a new class of antimicrobial agent against broad range microbial organisms and further biological studies and modification are under progress.

3. EXPERIMENTAL SECTION

3.1. Materials & Methods

Melting points of all the compounds were recorded on Casia-Siamia (VMP-AM) melting point apparatus and were uncorrected. IR spectra were recorded on a Perkin-Elmer FT-IR 240-C spectrophotometer using KBr optics. ¹H NMR spectra were recorded on Bruker AV 300 MHz and INOVA 400 & 500 MHz in DMSO-*d*₆ using TMS as internal standard. Multiplicity is reported as s (singlet), d (doublet),

t (triplet), quart (quartet), m (multiplet) and br (broad), respectively. All the reactions were monitored by thin layer chromatography (TLC) on pre-coated silica gel 60 F₂₅₄ (mesh); spots were visualized with UV light. Merck silica gel (60-120 mesh) was used for column chromatography CHN analysis was recorded on a vario EL analyzer.

3.2. Chemistry

Synthesis of 2-(1-(6-methyl/phenyl-4-(trifluoromethyl) Pyridin-2-yl) Piperidin-4-yl)-1*H*-benzo[d]imidazoles (**4a-h**).

General Procedure

A mixture of 2-chloro-6-phenyl/methyl-4-(trifluoromethyl) pyridine **3** (1 mmol) and 6-substituted-2-(piperidin-4-yl)-1*H*-benzo[d]imidazole (1.2 mmol) was charged and added NaHCO₃ (3.8 mmol), and DMF (5 mL) were added. The reaction mixture was heated for 18 h at 120-125 °C in N₂ atmosphere. Progress of the reaction was monitored by TLC. The reaction mixture was cooled and poured onto crushed ice. That obtained solid was filtered, washed with water, hot 4N HCl and followed by water. The obtained crude product was purified at the column chromatography over silica gel using the solvent system 4:6 ethyl acetate: hexane as eluent to give the title compounds (**4a-h**).

2-(1-(6-Phenyl-4-(trifluoromethyl) Pyridin-2-yl) Piperidin-4-yl)-1*H*-benzo[d]imidazole (**4a**)

Yield: 76 %. M.P: 202-204 °C, I.R: 3447.15 (-NH), 1608.96 (-C=N-), 1392.56 (CF₃). ¹H NMR (DMSO-*d*₆): δ 2.03-2.11 (m, 2H), 2.24 (d, 2H, *J* = 11.71 Hz), 3.16-3.22 (m, 3H), 4.60-4.65 (d, 2H, *J* = 12.84 Hz), 6.84 (s, 1H), 7.17-7.21 (m, 2H, Ar-H), 7.44-7.49 (m, 5H, Ar-H), 7.65 (s, 1H, Ar-H), 8.02 (d, 2H, Ar-H, *J* = 6.61Hz), 11.69 (br., NH). ESI-MS *m/z*: 423 (M⁺+H). Anal. Calcd. for C₂₄H₂₁F₃N₄: C, 68.24; H, 5.01; N, 13.26. Found: C, 68.19; H, 4.96; N, 13.25.

6-Chloro-2-(1-(6-phenyl-4-(trifluoromethyl) Pyridin-2-yl) piperidin-4-yl)-1*H*-benzo[d]imidazole (**4b**)

Yield: 73.4 %. M.P: >250 °C, I.R: 3436.87 (-NH), 1614.16 (-C=N-), 1416.48 (CF₃). ¹H NMR (DMSO-*d*₆): δ 2.03-2.19 (m, 2H), 2.33-2.44 (m, 2H), 3.69-3.82 (m, 3H), 4.68-4.73 (d, 2H, *J* = 13.59 Hz), 6.88 (s, 1H, Ar-H), 7.26 (s, 1H, Ar-H), 7.43-7.50 (m, 3H, Ar-H), 7.57 (m, 2H), 7.72 (d, 1H, Ar-H, *J* = 8.87 Hz), 8.03 (d, 2H, Ar-H, *J* = 7.55 Hz),

11.25 (br., NH). ESI-MS m/z : 457 ($M^+ + H$). Anal. Calcd. for $C_{24}H_{20}ClF_3N_4$: C, 63.09; H, 4.41; N, 12.26. Found: C, 63.05; H, 4.39; N, 12.24.

6-Fluoro-2-(1-(6-phenyl-4-(trifluoromethyl) pyridin-2-yl) piperidin-4-yl)-1H-benzo[d] Imidazole (4c)

Yield: 76 %. M.P: >250 °C, I.R: 3443.67 (-NH), 1612.70 (-C=N-), 1397.37 (C-F). 1H NMR (DMSO- d_6): δ 2.07-2.09 (m, 2H), 2.36-2.39 (d, 2H, J = 11.09 Hz), 3.19-3.25 (m, 2H), 3.73-3.79 (m, 1H), 4.70-4.73 (d, 2H, J = 13.16 Hz), 6.88 (s, 1H), 6.99 (d, 1H, J = 8.09 Hz), 7.22-7.28 (m, 3H, Ar-H), 7.31-7.35 (t, 1H, Ar-H, J = 8.09 Hz), 7.75 (t, 1H, Ar-H, J = 8.09 Hz), 8.02-8.04 (m, 3H, Ar-H), 11.61 (br., NH). ESI-MS m/z : 441 ($M^+ + H$). Anal. Calcd. for $C_{24}H_{20}F_4N_4$: C, 65.45; H, 4.58; N, 12.72. Found: C, 65.43; H, 4.56; N, 12.68.

6-Methyl-2-(1-(6-phenyl-4-(trifluoromethyl) pyridin-2-yl) piperidin-4-yl)-1H-benzo[d] Imidazole (4d)

Yield: 65 %. M.P: >250 °C, I.R: 3438.53 (-NH), 1612.43 (-C=N-), 1396.46 (CF_3). 1H NMR (DMSO- d_6): δ 2.00-2.11 (m, 2H), 2.27-2.31 (m, 2H), 2.47 (m, 3H, Ar-CH₃), 3.15-3.24 (m, 2H), 3.48-3.56 (m, 1H), 4.64-4.68 (d, 2H, J = 12.84 Hz), 6.85 (s, 1H, Ar-H), 7.12-7.15 (d, 1H, Ar-H, J = 8.50 Hz), 7.23 (s, 1H, Ar-H), 7.42-7.50 (m, 5H, Ar-H), 8.02-8.04 (d, 2H, Ar-H, J = 6.80 Hz), 11.32 (br., NH). ESI-MS m/z : 437 ($M^+ + H$). Anal. Calcd. for $C_{25}H_{23}F_3N_4$: C, 68.79; H, 5.31; N, 12.84. Found: C, 68.74; H, 5.29; N, 12.81.

Phenyl (2-(1-(6-phenyl-4-(trifluoromethyl) Pyridin-2-yl) piperidin-4-yl)-3H-benzo[d] Imidazol-5-yl) Methanone (4e)

Yield: 70 %. M.P: >250 °C, I.R: 3359.15 (-NH), 1612.81 (-C=N-), 1396.45 (CF_3). 1H NMR (DMSO- d_6): δ 2.11-2.16 (m, 2H), 2.39-2.43 (m, 3H), 3.74-3.86 (m, 2H), 4.71-4.75 (d, 2H, J = 13.40 Hz), 6.89 (s, 1H, Ar-H), 7.27 (s, 1H, Ar-H), 7.46-7.55 (m, 5H, Ar-H), 7.76-7.78 (d, 3H, Ar-H, J = 7.17 Hz), 7.89-7.90 (m, 2H, Ar-H), 8.03-8.05 (m, 2H, Ar-H), 8.17 (s, 1H, Ar-H), 11.12 (br., NH). ESI-MS m/z : 527 ($M^+ + H$). Anal. Calcd. for $C_{31}H_{25}F_3N_4O$: C, 70.71; H, 4.79; N, 10.64. Found: C, 70.67; H, 4.75; N, 10.61.

2-(1-(6-Methyl-4-(trifluoromethyl) Pyridin-2-yl) Piperidin-4-yl)-1H-benzo[d] Imidazole (4f)

Yield: 62 %. M.P: 185-187 °C, I.R: 3446.59 (-NH), 1612.70 (-C=N-), 1397.37 (CF_3). 1H NMR (DMSO- d_6): δ 1.25-1.31 (m, 2H), 1.97-2.07 (m, 3H), 2.17-2.22 (m, 2H), 2.41 (s, 3H, Ar-CH₃), 4.45-4.48 (d, 2H, J = 13.22 Hz), 6.65-6.72 (m, 2H, Ar-H), 7.40-7.44 (m, 2H, Ar-H), 7.56-7.64 (m, 2H), 11.54 (br., NH). ESI-MS m/z : 361 ($M^+ + H$). Anal. Calcd. for $C_{19}H_{19}F_3N_4$: C, 63.32; H, 5.31; N, 15.55. Found: C, 63.29; H, 5.28; N, 15.50.

(2-(1-(6-Methyl-4-(trifluoromethyl) Pyridin-2-yl) Piperidin-4-yl)-3H-benzo[d] Imidazol-5-yl), phenyl-methanone (4g)

Yield: 59 %. M.P: 192-194 °C, I.R: 3372.12 (-NH), 1612.79 (-C=N-), 1402.27 (CF_3). 1H NMR (DMSO- d_6): δ 1.97-2.10 (m, 2H), 2.20-2.23 (m, 2H), 2.44 (s, 3H, CH₃), 3.07-3.18 (m, 2H), 3.33-3.40 (m, 1H), 4.51-4.59 (d, 2H,

J = 13.97 Hz), 6.64 (s, 1H), 6.73 (s, 1H), 7.47-7.52 (m, 2H, Ar-H), 7.58-7.63 (m, 2H, Ar-H), 7.73-7.80 (m, 3H, Ar-H), 8.03 (s, 1H, Ar-H), 11.46 (br., NH). ESI-MS m/z : 465 ($M^+ + H$). Anal. Calcd. for $C_{26}H_{23}F_3N_4O$: C, 67.23; H, 4.99; N, 12.27; O, 12.06. Found: C, 67.19; H, 4.96; N, 12.04.

6-Chloro-2-(1-(6-methyl-4-) pyridin-2-yl) Piperidin-4-yl)-1H-benzo[d] Imidazole (4h)

Yield: 62 %. M.P: 197-198 °C, I.R: 3441.75 (-NH), 1612.99 (-C=N-), 1401.31 (CF_3). 1H NMR (DMSO- d_6): δ 1.96-2.04 (m, 2H), 2.19-2.24 (m, 2H), 2.45 (s, 3H, Ar-CH₃), 3.05-3.11 (m, 2H), 3.22-3.28 (m, 1H), 4.49-4.53 (d, 2H, J = 13.55 Hz), 6.64 (s, 1H), 6.67 (s, 1H), 7.16-7.19 (m, 1H, Ar-H), 7.47-7.49 (m, 1H, Ar-H), 7.53-7.55 (m, 1H, Ar-H), 11.23 (br., NH). ESI-MS m/z : 395 ($M^+ + H$, 100%). Anal. Calcd. for $C_{19}H_{18}ClF_3N_4$: C, 57.80; H, 4.60; N, 14.18. Found: C, 57.7; H, 4.55; N, 14.15.

Synthesis of N-alkylated 2-(1-(6-phenyl-4-(trifluoromethyl) pyridin-2-yl) piperidin-4-yl)-1H-benzo[d] Imidazoles (5a-c)

General Procedure:

A mixture of 2-(1-(6-Phenyl-4-(trifluoromethyl) pyridin-2-yl) piperidin-4-yl)-1H-benzo[d] imidazole (**4a**) (0.355 mmol), activated K_2CO_3 (1.78 mmol), alkyl bromide (3 equiv.) and acetone (3 mL) was refluxed in N_2 atmosphere for 18-24 h. Reaction progress was monitored by TLC. Acetone was evaporated from the reaction mixture and DM water was added to the residue. The solid obtained was filtered, washed with water and dried at a temperature of 60 °C. Crude product obtained was purified at the column chromatography over silica gel at 4:6 ethyl acetate: hexane as eluent.

1-Ethyl-2-(1-(6-phenyl-4-(trifluoromethyl) Pyridine-2-yl) piperidin-4-yl)-1H-benzo[d] Imidazole (5a)

Yield: 77 %. M.P: 136-138 °C, I.R: 1611.35 (-C=N-), 1393.35 (CF_3). 1H NMR (DMSO- d_6): δ 1.50-1.55 (t, 3H, N-CH₂-CH₃, J = 6.99 Hz), 2.09-2.30 (m, 4H), 3.14-3.27 (m, 3H), 4.30-4.38 (q, 2H, N-CH₂-CH₃, J = 6.99 Hz), 4.69-4.73 (d, 2H, J = 13.60 Hz), 6.84 (s, 1H, Ar-H), 7.24 (s, 1H, Ar-H), 7.30-7.33 (m, 2H, Ar-H), 7.40-7.48 (m, 5H, Ar-H), 8.02 (d, 2H, Ar-H, J = 6.98 Hz). ESI-MS m/z : 451 ($M^+ + H$). Anal. Calcd. for $C_{26}H_{25}F_3N_4$: C, 69.32; H, 5.59; N, 12.44. Found: C, 69.28; H, 5.54; N, 12.39.

2-(1-(6-Phenyl-4-(trifluoromethyl) Pyridin-2-yl) piperidin-4-yl)-1-propyl-1H-benzo[d] Imidazole (5b)

Yield: 75 %. M.P: 150-152 °C, I.R: 1612.61 (-C=N-), 1396.91 (CF_3). 1H NMR (DMSO- d_6): δ 1.01-1.06 (t, 3H, N-CH₂-CH₂-CH₃, J = 7.55 Hz, 7.18 Hz), 1.87-1.94 (m, 2H, N-CH₂-CH₂-CH₃, J = 7.55 Hz, 7.18 Hz), 2.05-2.23 (m, 5H), 3.18-3.25 (m, 2H), 4.17-4.22 (m, 2H, N-CH₂-CH₂-CH₃, J = 7.55 Hz, 7.18 Hz), 4.67-4.72 (d, 2H, J = 13.21 Hz), 6.85 (s, 1H, Ar-H), 7.18-7.24 (m, 1H, Ar-H), 7.36-7.48 (m, 5H, Ar-H), 7.57 (s, 1H, Ar-H), 7.69 (d, 1H, Ar-H, J = 7.74 Hz), 8.02 (d, 2H, Ar-H, J = 6.61 Hz). ESI-MS m/z : 465 ($M^+ + H$). Anal.

Calcd. for $C_{27}H_{27}F_3N_4$: C, 69.81; H, 5.86; N, 12.06. Found: C, 69.78; H, 5.82; N, 12.02.

2-(2-(1-(6-Phenyl-4-(trifluoromethyl) Pyridin-2-yl) Piperidin-4-yl)-1H-benzof[d]imidazol-1-yl) Acetonitrile (5c)

Yield: 83 %. M.P: 174-176 °C, I.R: 1610.87 (C=N-), 1396.06 (CF_3). 1H NMR (DMSO- d_6): δ 2.18-2.27 (m, 4H), 3.23-3.37 (m, 2H), 3.49-3.62 (m, 1H), 4.71-4.80 (d, 2H, J = 11.51 Hz), 5.59 (s, 2H, N-CH₂-CN), 6.97 (s, 1H, Ar-H), 7.27-7.30 (m, 2H, Ar-H), 7.41-7.49 (m, 3H, Ar-H), 7.57-7.61 (m, 1H, Ar-H), 7.65-7.69 (m, 1H, Ar-H), 8.08-8.10 (m, 2H, Ar-H). ESI-MS m/z : 462 (M^+ +H). Anal. Calcd. for $C_{26}H_{22}F_3N_5$: C, 67.67; H, 4.81; F, 12.35; N, 15.18. Found: C, 67.63; H, 4.77; N, 15.14.

2-(Piperidin-4-yl)-1-propyl-1H-benzof[d]imidazole (8)

Yield: 72%. Gummy, I.R: 1608.96 (C=N-), 1392.56. 1H NMR (DMSO- d_6): δ 0.94-1.05 (m, 4H), 1.65-1.75 (m, 1H), 1.83-2.10 (m, 4H), 3.01-3.09 (m, 2H), 3.28-3.32 (d, 2H, J = 12.84 Hz), 4.04-4.12 (m, 2H), 6.69 (d, 1H, Ar-H, J = 9.01 Hz), 7.20-7.35 (m, 2H, Ar-H), 7.67-7.6 (d, 1H, Ar-H, J = 8.56 Hz). ESI-MS m/z : 244 (M^+ +H). Anal. Calcd. for $C_{15}H_{21}N_3$: C, 74.03; H, 8.70; N, 17.27; Found: C, 74.69; H, 8.67; N, 17.24.

3.3. In vitro Antimicrobial Assay

Antibacterial Activity: Procedure

Minimum inhibitory concentration (MIC) assay of hybrid molecules **4a-h** and **5a-c** was done according to CLSI Standard Protocol by the micro dilution method in liquid medium distributed in 96-well plates. In each test a microbial culture control and a sterility control (negative) were performed. The plates were incubated for 24 hours at 37 °C. The lowest concentration which inhibited the growth of test organism was considered as MIC (μ g/mL) value for the tested compound. Norfloxacin and Penicillin were used as standard drugs. The minimum inhibitory concentration (MIC) values are presented in the Table 1.

Antifungal Activity: Procedure

In vitro antifungal activity of hybrid molecules **4a-h** and **5a-c** was studied against the fungal strains. The Potato Dextrose Agar (PDA) medium was suspended in distilled water (39 g in 1000 mL) and heated to boiling until it dissolved completely, the medium and Petri dishes were autoclaved at pressure of 15 lb/inc² for 20 min. Agar well bioassay was employed for testing antifungal activity. The medium was poured into sterile Petri dishes under aseptic conditions in a laminar air flow chamber. When the medium in the plates solidified, 0.5 mL of (week old) culture of test organism was inoculated and uniformly spread over the agar surface with a sterile L-shaped rod. Different concentrations of solutions were prepared by dissolving the compound in DMSO. After inoculation, wells were scooped out with 6 mm sterile cork borer and the lids of the dishes were replaced. To each well, different concentrations (100 μ g/mL and 150 μ g/mL) of test solutions were added. Controls were maintained. The

plates were incubated at 27 °C for 48 h. Inhibition zones were measured, the diameter was calculated in millimeter and the corresponding results were in Table 2.

4. CONCLUSION

In conclusion, we have discovered that a number of benzimidazole-pyridine-piperidine hybrids displayed moderate activity against various fungal organisms as well as bacterial organisms. One promising compound **5a** exhibited broad spectrum antifungal activity including antibacterial activity *in vitro* and was more potent than the standard drugs.

CONFLICT OF INTEREST

The authors confirm that this article content has no conflict of interest.

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